

Double Integrals

ANSWER KEY



Double integrals over a rectangle

- 1 Set up the double integral of $f(x, y) = xy$ over the rectangle $[0, 2] \times [0, 3]$ (integrate y first).
$$\int_0^2 \int_0^3 xy \, dy \, dx.$$

Iterated integrals

- 2 Evaluate $\int_0^2 \int_0^3 xy \, dy \, dx$.
Inner: $\int_0^3 xy \, dy = x \cdot \frac{y^2}{2} \Big|_0^3 = \frac{9x}{2}$. Outer: $\int_0^2 \frac{9x}{2} \, dx = \frac{9}{4} x^2 \Big|_0^2 = 9$.
- 3 Evaluate $\int_0^1 \int_0^2 (x^2 + y) \, dy \, dx$.
Inner: $\int_0^2 (x^2 + y) \, dy = x^2 y + \frac{y^2}{2} \Big|_0^2 = 2x^2 + 2$. Outer: $\int_0^1 (2x^2 + 2) \, dx = \frac{2}{3} x^3 + 2x \Big|_0^1 = \frac{8}{3}$.

Fubini's Theorem: order of integration

- 4 Verify Fubini's Theorem for $f(x, y) = xy$ on $[0, 1] \times [0, 2]$ by evaluating the integral in both orders.
Order $dy \, dx$: inner $= x \cdot \frac{y^2}{2} \Big|_0^2 = 2x$, outer $= \int_0^1 2x \, dx = 1$. Order $dx \, dy$: inner $= y \cdot \frac{x^2}{2} \Big|_0^1 = \frac{y}{2}$, outer $= \int_0^2 \frac{y}{2} \, dy = 1$. Both orders give 1.

Double integrals over general regions

- 5 Evaluate $\int_0^1 \int_0^x (x + y) \, dy \, dx$.
Inner: $\int_0^x (x + y) \, dy = xy + \frac{y^2}{2} \Big|_0^x = x^2 + \frac{x^2}{2} = \frac{3x^2}{2}$. Outer: $\int_0^1 \frac{3x^2}{2} \, dx = \frac{1}{2} x^3 \Big|_0^1 = \frac{1}{2}$.

Polar coordinates

- 6 Convert $\iint_D (x^2 + y^2) \, dA$ to polar coordinates, where D is the disk $x^2 + y^2 \leq 4$, and evaluate.
 $x^2 + y^2 = r^2$, $dA = r \, dr \, d\theta$. $\int_0^{2\pi} \int_0^2 r^3 \, dr \, d\theta$. Inner: $\frac{r^4}{4} \Big|_0^2 = 4$. Outer: $\int_0^{2\pi} 4 \, d\theta = 8\pi$.
- 7 Evaluate $\iint_D xy \, dA$ where D is the quarter disk $x^2 + y^2 \leq 1$, $x \geq 0$, $y \geq 0$, using polar coordinates.
 $xy = r^2 \sin \theta \cos \theta$, $dA = r \, dr \, d\theta$, $\theta \in [0, \frac{\pi}{2}]$, $r \in [0, 1]$. $\int_0^{\pi/2} \sin \theta \cos \theta \, d\theta \cdot \int_0^1 r^3 \, dr = \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}$.